

The view from ultramiminalism

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- ▶ No $\rightarrow, \Pi$
- ▶ Developed as tool for working with locales and toposes
- ▶ Foundational role — for all maths??? Is it truly constructive?

Beyond the minimalist foundations

Sambin's "Positive Topology" [Sam25, p.112]:

This is why the resulting foundation, corresponding to the kind of constructivism adopted in this book, has been called minimalist. An advantage of this approach is that all readers can follow the text without giving up their own foundational attitude.

Not quite true! Can still reject connectives.

Two *ultraminimalist* foundations — reject $\neg$, $\rightarrow$, Π .

- ▶ *Geometric* mathematics for Grothendieck toposes:
finite limits, arbitrary colimits
- ▶ *Arithmetic* mathematics for arithmetic universes (AUs):
finite limits, finite colimits, parametrized list objects
Still captures important infinities.

From topology to ultraminimalism

For propositional geometric logic – locales

$\cap$ and $\cup$ of open sets become $\wedge$ and $\vee$

$\neg$ and $\rightarrow$ exist in Heyting algebra of open sets (by taking interior), but are not preserved by inverse image.

For first order geometric logic – toposes

Opens not enough – must use *sheaves* (= local homeomorphisms)

Geometric constructions of sheaves are computed fibrewise.

For arithmetic universes

Propositional logic rejects infinitary disjunction.

First order logic with arithmetic sort constructors captures some $\forall$ s using $\exists$

– sufficient for practical purposes, including real analysis.

Current status of theory

Geometric

- ▶ Infinite disjunctions not truly ultraminimalist
 - but still serve to motivate ultraminimalist concerns
- ▶ Solid theory [Joh02, B4]
- ▶ Displays the intrinsic topology.

Arithmetic

- ▶ Truly ultraminimalist objective
- ▶ Theory being ported from toposes
 - but still under construction [MV12, Vic19, Vic17, Des25] ...

Hybrid – Grothendieck toposes with arithmetic methods

- ▶ Effective tool for point-free reasoning [Vic99] ... [NV22]
- ▶ Shows considerable range for arithmetic methods

Issues

Ultraminimalism natural setting for topology and continuity

Discontinuity introduced only with negative constructions.

Ultraminimalism as tool for locales, toposes.

Different ontologies: constructive v. observational

Ultraminimalism, and topology, are observational.

Denotational semantics: Programmer constructs, user observes.

Roles can alternate.

Concrete v. abstract

Sets, classes, collections ... *spaces* ...

— must be reexamined in the presence of topology.

Internal and external AUs

Construction appears in observational ontology?

Basic tricks of point-free topology

Very general!

Really trick of *classifying categories*.

Important when logic is incomplete (eg geometric).

Restores pointwise reasoning.

Point-free topology trick 1: spaces as theories

Point of space = model of a logical theory

- ▶ No single collection of points: models can be found in any suitable category.
- ▶ Models are transported by suitable functors.

Let us write $\mathfrak{Suit}^{op}$ for a (2-)category of suitable categories and functors.

Examples

- ▶ *Grothendieck toposes* – suitable functors preserve colimits, finite limits
- ▶ Grothendieck toposes *relative to base topos* $\mathcal{S}$ – suitable functors are inverse image parts f^* of geometric morphisms. $\mathfrak{Suit} \simeq \mathfrak{BT}op/\mathcal{S}$ [Joh02, B4].
- ▶ Categories are *Joyal's arithmetic universes* (AUs) [Mai10], functors are AU-functors (non-strict)

Point-free topology trick 2: spaces as representing objects

Full semantics of theory is $\mathbb{S}\text{uit}$ -indexed category, $\mathcal{E} \mapsto \mathbb{T}(\mathcal{E})$

$\mathbb{T}(\mathcal{E})$ describes category of models at every suitable $\mathcal{E}$.

Reindexing describes model transport by suitable functors.

Massive structure!

Representing object $(\mathcal{E}_{\mathbb{T}}, M_{\mathbb{T}})$

Suitable category $\mathcal{E}_{\mathbb{T}}$ with *generic model* $M_{\mathbb{T}}$ in $\mathbb{T}(\mathcal{E}_{\mathbb{T}})$, which can be transported to any specific model M in $\mathbb{T}(\mathcal{F})$ by a unique suitable functor.

Then $\mathbb{T}(\mathcal{F}) \simeq \mathbb{S}\text{uit}(\mathcal{F}, \mathcal{E}_{\mathbb{T}})$.

$(\mathcal{E}_{\mathbb{T}}, M_{\mathbb{T}})$ represents whole of theory $\mathbb{T}$.

Do representing objects exist?

For toposes and geometric theories? Yes! – classifying toposes.

For AUs? We know many do. Conjecture this is general.

Write $[\mathbb{T}]$ for $\mathcal{E}_{\mathbb{T}}$ as space, point = model

Model in $\mathcal{F}$ = (generalized) point at $\mathcal{F}$ = $\text{map } \mathcal{F} \rightarrow \mathcal{E}_{\mathbb{T}}$

Point-free topology trick 3: maps constructed pointwise

Map $f: [\mathbb{T}] \rightarrow [\mathbb{T}']$ – opposite of suitable functor

Let M be a point of $[\mathbb{T}]$ (model of $\mathbb{T}$)

... work “suitably” ...

Construct point $f(M)$ of $[\mathbb{T}']$.

Inside box = internal mathematics of $\mathcal{E}_{\mathbb{T}}$, with $M = M_{\mathbb{T}}$.

Working “suitably”

Constructions available in every suitable category, and preserved by every suitable functor.

Transform every $\mathbb{T}$ model to a $\mathbb{T}'$ -model

$f(M)$ is model of $\mathbb{T}'$ in $\mathcal{E}_{\mathbb{T}}$, hence $\mathfrak{S}\text{uit}$ morphism $f: [\mathbb{T}] \rightarrow [\mathbb{T}']$.

Construction f transports to *any* model of $\mathbb{T}$, $M: \mathcal{F} \rightarrow \mathcal{E}_{\mathbb{T}}$.

It works by composition in $\mathfrak{S}\text{uit}$, $M \mapsto f \circ M$.

Ultraminimalism as tool for locales and toposes

Define locales and toposes as *theories*

eg formal topologies, sites

No need to calculate lattices of opens or categories of sheaves.

Define maps, geometric morphisms by their action on *points*

Always enough points – generic point suffices.

Use free algebra constructions instead of “arbitrary” colimits

Still geometric, but don't depend on base topos.

Also suitable for arithmetic universes.

General idea set out in “Topical categories of domains” [Vic99].

See [Vic07] for example of Dedekind reals.

Ultraminimalist foundations?

- ▶ *Observational* ontology
- ▶ How to relate it to constructive ontology?
- ▶ Observations are prior: constructions as particular case.
- ▶ Deal with observability issues first (with topology!), then constructivist issues.
- ▶ Everything looks different, unfamiliar
 - “throwing away one’s favourite toys” (Benno van den Berg)

Scott-Strachey denotational semantics [Sto77]

Syntax $\longrightarrow$ semantics, $C \mapsto \llbracket C \rrbracket$

C is an element of a set of program fragments

$\llbracket C \rrbracket$ is a point of a semantic domain, a *topological space* of meanings.

What is the role of the topology? (Abramsky, Smyth, Vickers)

Technically: cuts down the size of the function space, because only interested in continuous functions.

Ontologically: topology expresses *observational power* of the user.¹

Programmer: can inspect the program code.

User: can only interact with the running program.

Two different viewpoints

Programmer's: what a construction is.

User's: what it achieves.

¹Note: User view, expressed axiomatically, cannot rule out non-constructible points, even if they are believed not to exist.

Related examples

Application Program Interface (API)

Important firewalling principle of software engineering! Decouples layers of construction.

Application program *uses* a *preprogrammed* software library.

Doesn't inspect library code, but works through API specification.

Library can be revised without affecting user.

Relationship can be chained – application program might be another library.

Concrete v. abstract mathematics

Axiomatization in mathematics is similar.

eg abstract groups instead of concrete groups of permutations or matrices.

Ultraminimalism as observational mathematics

Connectives $\wedge, \vee$ have direct observational interpretation

– unlike $\neg, \rightarrow$. [Vic89]

For first order [Vic10] —

A *set* (semantically) is defined by:

- ▶ what is needed to observe (“apprehend”) elements, and
- ▶ what is needed to observe two elements are equal.

Finite limits and arbitrary colimits justifiable in terms of these recipes.

Observability is key issue here

cf. Sambin’s “collections”.

Geometric problem: infinite disjunctions/colimits

Need theory of indexing sets. (Geometrically: use base topos $\mathcal{S}$.)

Arithmetic base-independent solution

Use arithmetic sort constructors in internal maths.

Then eg $\bigvee_{n \in \mathbb{N}} \phi_n$ becomes $\exists n : \mathbb{N}. \phi(n)$.

Concrete sets, seem the same as Sambin's *sets*.

Constructions as observations

- ▶ To apprehend an element: construct a term.
- ▶ To observe equality: find a proof of equality.

Ultramini­malist theories

Positive connectives $\wedge, \vee, \exists, =$ used to build formulae.

Theory axioms are sequents $\phi \vdash_{xyz} \psi$

If you apprehend elements x, y, z and observe ϕ , then it's also possible to observe ψ .

Sequents are not observable! They are like scientific hypotheses.

Negation $\neg\phi$ is Popperian

If your theory with observations so far proves $\phi \vdash_{xyz} \perp$, and you observe ϕ , then the theory is wrong.

Spaces

- ▶ Everything topologized, general notion of “collection” is *space*.
- ▶ “Set” is topologically *discrete* space.
- ▶ Some sets are more concrete – like sets of [Sam25].

Set = discrete space

- ▶ Characterized topologically [JT84] by maps $X \rightarrow 1$ and $\Delta: X \rightarrow X \times X$ being *open*.
- ▶ Translates observationally² to apprehension and equating of elements [Vic10].
- ▶ Observational recipes can be combined in finitary ways to construct new sets, eg list object $\text{List } X$, finite powerset $\mathcal{F}X$.
- ▶ Still need to deal with sets prior to spaces, but now they are ultramiminalistic and objects of AUs.³
- ▶ In appearance the same as “collection” in [Sam25, pp.64, 116], but the requirement for equality to be *open* (observable) is an important part of the ultramiminalist analysis and excludes most collections in [Sam25].
- ▶ Example: set of cats in Venice

²more or less?

³Conservativity conjecture here: discrete spaces à la [JT84] work the same way as observational sets.

Concrete set = set as in [Sam25]

- ▶ To apprehend: perform a construction
To equate: find a proof of equality
- ▶ Generally sets constructed “out of nothing”
eg $\emptyset$, 1 , $\mathbb{N} = \text{List } 1$, $\mathbb{Q}$.
Not $\mathbb{R}$ – it’s a space.
- ▶ Equality need not be decidable – eg if undecidable word problem.

General spaces

- ▶ *Localic* if corresponding theory is essentially propositional.
- ▶ *Specialization* always present.
For a localic space it's a partial order.
In general it's homomorphisms between models
– corresponding to the 2-cells between geometric morphisms
Filtered colimits of points always exist, and are preserved by maps.
- ▶ The topology is not negotiable.
You cannot ultraminimalistically extract “the set of points”
(the discrete coreflection in elementary topos theory)
– it's not preserved by the inverse image functors.
Topologically, that would be trying to approximate a general bundle by a local homeomorphism, and the approximation is too crude to work successfully.

Examples of spaces

Exponentiation Y^X not generally available

If X, Y are sets: Y^X exists *but is not in general a set*.

Equality between functions not observable; topology is point-open.

Power “set” $\mathcal{P}X$ not a set

Specialization order is subset inclusion.

$\mathcal{P}X$ is ideal completion $\text{Idl}(\mathcal{F}X)$ and topology is Scott.

The real line $\mathbb{R}$

That's the Dedekind reals, with Euclidean topology.

For other versions, the topology adjusts for the change in structure.

One-sided reals (topologies of semicontinuity) are important.

The space of sets (and similarly other “proper classes”)

Theory has one sort and nothing else.

Representing object is object classifier – for AUs see [Des25].

Specialization morphisms are functions.

External – internal AUs

An AU can have internal AUs.

(Doesn't work for Grothendieck toposes!)

Speculation: External:Internal :: Observational:Constructive

External AU = ambient observational maths

Internal AU = constructed maths

Can we see constructivist issues living at this interface??

Nesting of levels needs better understanding

Related to Gödel's theorems (à la Joyal)?

May be needed to provide proper discussion of representing objects.

cf. [Des25] use of internal AUs and indexed AUs.

Conclusions

- ▶ *Methodologically*: Easy to overlook the impact of choice of connectives.
- ▶ *Logically*: Ultraminimalism prior to systems with negatives
- ▶ *Philosophically*: Observations prior to constructions?
- ▶ *Mathematically*: There's a rich mathematics despite the apparent logical weakness.
- ▶ *Engineering*: Topology much more elegant ultraminimally. Much easier than “unscrambling the omelette” after negative connectives have introduced discontinuities.
- ▶ *Research*: Still much to be understood about AUs.

Learning to ride a bike

Add stabilizers



Learning to ride a bike: Better solution by subtraction

Subtract pedals



Focus on learning to balance.

Pedals = negative connectives

Stabilizers = topological bureaucracy

Add stabilizers



Not a bike!

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